



Buddy - A Personal Companion to Match People with Cognitive Disabilities and AT

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Abstract. This paper presents a novel web-based repository and recommender system to match the requirements of users with cognitive disabilities with available AT solutions when working on the web. Based on personal data from users (questionnaire, simple games) stored in a profile and tool data provided by AT solution providers and AT vendors, Buddy recommends tools supporting particular cognitive needs of target users. Buddy intends to bridge the gap between the growing amount of AT solutions and the actual low level of uptake by involving all stakeholder groups in the value chain on the Buddy platform.

Keywords: Web accessibility · Cognitive accessibility · Assistive technologies · Recommender system

1 Introduction

Buddy is a R&D effort addressing the need to overcome the gap between the growing number of Assistive Technology (AT) available for people with cognitive disabilities when working on the web and the actual low level of uptake and use in practice [10].

Based on an analysis of the manifold reasons for low uptake (e.g. findability for independent access to tools and features, lacking customization and personalization, missing profile building and matching process, low incentive and missing training of the support/care environment, low/no interest in innovation at funding/commissioning/administrating level e.g. [2, 11]), this paper presents a new and innovative web based approach to support the matching process of people with cognitive disabilities and AT. R&D lead to the development of *Buddy* providing:

1. A web **repository** to search, find, explore and exchange Assistive Technology (AT) more independently
2. An AI-based **recommender system** matching the user profile with parameters and functionalities of ATs. The recommender uses data provided by
 - users through answering questions and/or by playing a series of games to define the need of support for the different dimensions of cognitive skills

- AT solution providers adding and categorizing their tools for the matching process
 - An explicit user-item rating mechanism feeding data into the recommender as well as providing an exchange/discussion platform on AT use.
3. A **quality assurance** back-end workflow to ensure that AT entries stored in the system are up-to-date and of relevance for target users.

Buddy is intended as a personal companion assisting users with cognitive disabilities and their support environment in finding the right AT solution for a successful web experience. It provides a new level of contact to users and a unique, rich and growing source for requirements elicitation for AT designers, developers, providers, commissioner and funders to better match ATs and functionalities towards needs of users and to support personalization.

2 End User Survey

Preparatory desktop research and online user interviews were conducted before designing *Buddy* in order to determine how persons with cognitive disabilities find, retrieve, and use AT, and to assess needs and problems in these processes. All in all, 88 individual responses to the online survey were gathered and analysed. 65 responses corresponded to Swedish participants, 20 answers came from Austria, 2 from the United Kingdom, and the remaining response came from Lithuania¹.

Participants where first queried regarding whether they were responding the questionnaire for themselves or for a third person (e.g. a caregiver responding on behalf of a disabled individual). As shown in Fig. 1, approximately half of the respondents reported answering for themselves, whereas the second half required some form of assistance in filling in the questionnaire. Therefore, the gathered responses are likely to cover a wide range of the cognitive spectrum.

Table 1. Usage of AT by questionnaire participants (answer counts).

Responding for:	Myself	A friend or relative	A person I support	Other	N/A	Total
Uses AT	15	5	4	2	2	28
Does not use AT	27	25	6	2	0	60
Total	42	30	10	4	2	88

Participants were additionally inquired about their current usage of AT. The collected answers are summarized in Table 1. These results suggest that not

¹ The full end user survey report is part of Buddy's "Report on User and Technical Requirements", publicly available at <https://www.buddyproject.eu/main-results>.

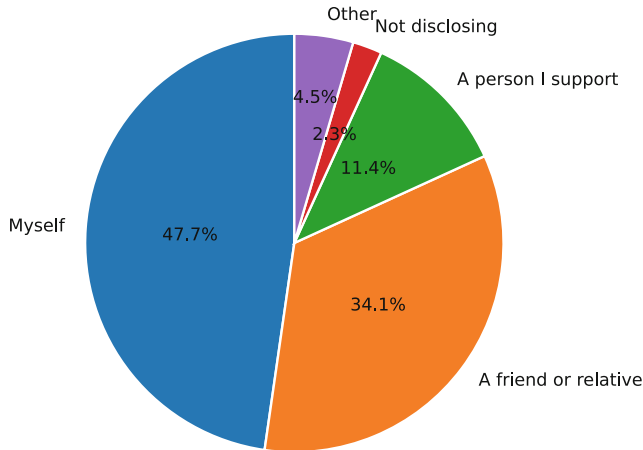


Fig. 1. Responses given by participants to the query “I am responding for...”, as percentage of total gathered answers (88).

employing AT results in limited autonomy for the person (only 45% answered for themselves).

Regarding the type of AT employed, reading support tools were used by the majority of users (64%), followed by AT to support writing (43%). AT usage by support category is followed by understanding (36%), memory (32%), focus (29%), managing time (29%), calculation (25%), managing tasks (21%), and managing choices (29%). In addition, most respondents (71%) reported using more than one AT type. These results suggest that users commonly employ a combination of AT; consequently, potential users of AT are likely to find a centralized repository of AT helpful for finding the right solution for their individual needs.

Next, questions were posed on aspects related to finding, obtaining, and using AT. Participants were asked about the source of the AT they use. 54% of them reported that they downloaded a tool from the internet for free. 35% reported buying tools themselves, and only 30% obtained AT from a government agency, and/or from their school, employer, or similar. When asked whether it was easy to find AT that meets their needs, only one third of participants reported being able to find it without external help (more precisely, 15% reported having no problem at all to find AT whereas 18% stated that the process of finding AT could be improved despite managing it by themselves); on the other hand, 30% declared not being able to find a better tool, and 29% stated that they could find suitable AT only with help. Again, these results align with the assumption that a majority of users would benefit from a centralized intelligent repository that helps them in finding suitable AT by themselves.

In addition, 25% of users reported needing help in using AT even if they found a good solution for their needs. Help may be necessary in setting up tool preferences, installing the tool, learning how to use it, getting updates, and

during regular use of the tool. These results suggest that there exists a significant gap in the AT provision process when it comes to training, support, and maintenance. Users did not only report needing external help when employing AT, but also expressed that current solutions ought to be improved - only 25% of surveyed users said that their current AT works flawlessly. Suggested areas of improvement include ease of use, more settings/preferences, need for better support such as training, and additional functionality.

Participants who reported not using AT were asked the reasons why they forgo it. 30% of them said that they simply do not need it. However, the majority reported not being able to find the right tool for their needs: 29% said not to know where to find AT, and 27% reported that they could not find AT that works for them. In addition, 5% said that it was too difficult to get the solutions they found.

In short, despite our sample size ($N = 88$) not being statistically significant, this preparatory research provides useful insights with regards to the barriers currently present in the AT provision process for persons with cognitive disabilities. The gathered answers suggest that there is an important need for a centralized repository with easily available and clear information about AT supporting cognitive user needs, including installation and usage instructions as well as clear classification and search functionality for tools according to the support categories they cover. Such a repository would also help mitigate the existing latency in regard to user needs, by letting users autonomously find AT for their specific support needs among the myriad of existing solutions. In turn, AT providers would also benefit from it by exposing their solutions to a large audience of potential users.

3 The Buddy Approach

The state-of-the-art analysis and a user-centered research, design and development approach based on the IPAR-UCD (Inclusive Participatory Action Research for User Centered Design) method [3] for including end-users as co-researchers, allowed us to specify, design and implement the concept and the components of Buddy. In addition, the different versions of the prototype and functionalities, developed in an cyclic agile approach, got evaluated by 18 end users with cognitive disabilities and from the neuro-diverse spectrum using the Think Aloud Protocol [9] and the Heuristic Evaluation [12] method, adapted to the requirements of the end users. Each component of Buddy is described next.

3.1 Accessible Web Repository of AT

Buddy intends to become a “one stop shop” for accessing ATs whose functionality hinges on a fully accessible Web application that serves as a central repository of individual AT tools and solutions enabling²:

² <https://develop.buddyproject.eu/>.

- Target users with a cognitive disability to search, (automatically) find, read about, download, and give feedback on AT solutions suitable for their individual support needs. This functionality is also useful for formal and informal care/support staff, teachers, trainers, employers, etc.; as well as administration and funding or commissioning bodies in search for AT.
- AT providers to add, describe, exchange, and get first hand feedback on their tools and solutions. Buddy invites AT solution providers but also other stakeholder to enter and categorize ATs into the public repository thereby exposing them to a suitable audience, see previous point.

The platform provides first hand access to end users, a communication and cooperation platform which is a rich source for AT providers to develop their tools further but also their market relations and cooperation. The cooperation with end user organizations, service provider organizations, administrative and political level will help to overcome reluctance and fears and should make Buddy the platform to be for the whole sector.

3.2 Profile Building via Traditional Forms and Gamification

To generate suitable AT recommendations for a specific user, the needs and preferences of that user must be known. In web accessibility, the key reference point for user needs are the standards of accessibility, namely, the standard EN 301 549 [5] and WCAG 2.1 [14]. However, none of these standards were conceived with the aim of proactively supporting cognitive user needs.

On the other hand, there exist guidelines issued by standardization agencies that do define cognitive accessibility. We have identified two such guidelines that had the most detailed definitions of cognitive user needs: namely [4] and [8]. To establish a relevant list of cognitive user needs to use as a basis for eliciting the user requirements, we mapped the definition of cognitive functions and needs in these guidelines against the cognitive abilities as defined in the international classification of functioning, disability and health [13] (the benchmark for defining human abilities and functions).

The list was also complemented by definitions of cognitive user requirements developed within research initiatives that explicitly look at requirements for the web, most notably the research conducted by the W3C COGA group [15]. The result was the following list of user needs for support that cover all required aspects:

- Reading
- Writing
- Understanding
- Calculation
- Focusing on a task or information, and keeping the focus
- Managing tasks (getting started and completing them)
- Memory
- Managing time (planning, allocating and controlling)

- Managing choices (evaluating options, deciding)

These preferences can be set by the user with a classic multi step web-form, where each step represents a user need mentioned above. However, previous projects and user involvement activities have shown that people with cognitive disabilities often struggle with long forms as they tend to be tedious and too complex. In addition, sometimes users do not know or are unable to express which types of support they need.

Therefore, a new innovative game-based approach [6] was implemented, in which users play mini games that aim to detect the users' needs for support. The games implemented address the cognitive dimensions/skills mentioned above. At the moment, 6 mini games have been developed covering 7 out of the 9 support categories. For further information on the gamification of user profile creation in Buddy, the reader is directed to [7].

With this gamification approach the capabilities of the user can be detected and added to his or her profile. The system allows and invites to do R&D for more and in particular more specific as well as attractive profiling games. Games have been developed and tested using the IPAR-UCD [3] approach to make sure that a high level of accessibility and usability is reached. The promising results invite to do more R&D to improve the approach.

3.3 AI Based Recommendations

Buddy uses both user profiles and AT entries stored in the repository as data sources for an intelligent recommender subsystem. Its main purpose is to find suitable ATs for specific users that may not be aware of their existence. In this manner, users are encouraged to try out new technologies that support their specific needs, thereby benefiting both users and AT vendors. The implemented hybrid recommender system hinges on two complementary methods:

- A knowledge-based recommendation approach that matches ATs to users directly by exploiting explicit knowledge about the support needs of users and support categories of ATs in the repository. A similarity score between ATs and the target user is computed, and the highest-scoring ATs may be recommended to the user.
- Data-driven recommendations that utilize user ratings of individual ATs to discover similar users and ATs regardless of their specific profiles. This system is inspired by well-established collaborative filtering methods commonly employed in e-commerce websites. Users are represented as a collection of score vectors, and similar ones to the target user are retrieved according to their similarity (Pearson's r correlation) in rating data-space. In this manner, suitable ATs for a target user may be found even if partial or incorrect knowledge about them is stored in the repository.

The final recommendations offered to a target user are based on a weighted mean of both scores in order to smooth the final score while benefiting from both

techniques. Initially, more importance is given to the knowledge-based score. As more user ratings are inputted into the system, emphasis is placed on the data-driven scores. Specifically, the final likeness score $s(a_i, u_t)$ for target user u_t and a target AT entry a_i (yet unrated by u_t) is computed as follows:

$$s(a_i, u_t) = w \cdot d(a_i, u_t) + (1 - w) \cdot k(a_i, u_t) \quad (1)$$

where d is the data-driven score, k is the knowledge-based score, and $w \in [0, 1]$ is a dynamic weighting term that corresponds to the system's confidence on the accuracy of the data-driven score i.e. the more users have been considered in the computation of d , the higher the value of w , up to a value of 1.0 once 100 users have been taken into account.

This method is an instance of a distributed weighted hybrid recommender system [1]. Candidate AT entries a_i for user u_t are sorted according to their value of $s(a_i, u_t)$. The sorted elements are then offered as suggestions to the target user, in descending order.

3.4 Quality Assurance (QA) Back-End

Lastly, to guarantee the highest quality of the system, a web back-end is provided, which allows control of AT descriptions and categorization by expert users. A moderation process ensures that each AT entry available to end users in Buddy meets the necessary quality and relevance criteria.

Users with *AT moderator* privileges are informed each time a new AT entry has been added by an *AT vendor* user to the system. They may then proceed to review the entry (appropriateness of the solution, quality and ease of reading of its description, proper categorization, correct download links, etc.) following internal guidelines. Once the tool has been deemed appropriate for its inclusion to the repository, a new revision is published and made available to all end users, which may search for it manually or get it automatically recommended if the solution matches their specific needs (c.f. Sect. 3.3).

Authors may edit their AT entries at any time, whereupon a new revision for the AT entry is created in the system. This new revision needs to go again through the QA workflow previously described. In this manner, Buddy allows AT vendors and providers to keep their solutions up to date in the repository while ensuring the quality of updated entries.

4 Conclusion

This paper presents a first prototype of an online platform that is able to recommend suitable AT to individual users with cognitive disabilities. The system intends to provide benefits to all stakeholder groups and to bridge the gap between potential and actual use of AT. It is intended and negotiated that the system is taken up by larger umbrella organizations to become sustainable and used at broad scale. It also provides a rich source for R&D at all levels of the system (e.g. user data for profiling, automatic/supported AT integration, AI based

recommendation, platform features). Extension to other target groups is on the agenda for future R&D.

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